

Project Design Phase Solution Architecture			
Date	31 August 2026	Team ID / Scope	x_1959081_emp_issue
Project Name	Employee Raise Issue Integration	Maximum Marks	4 Marks

SOLUTION ARCHITECTURE

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

EXAMPLE - SOLUTION ARCHITECTURE DIAGRAM

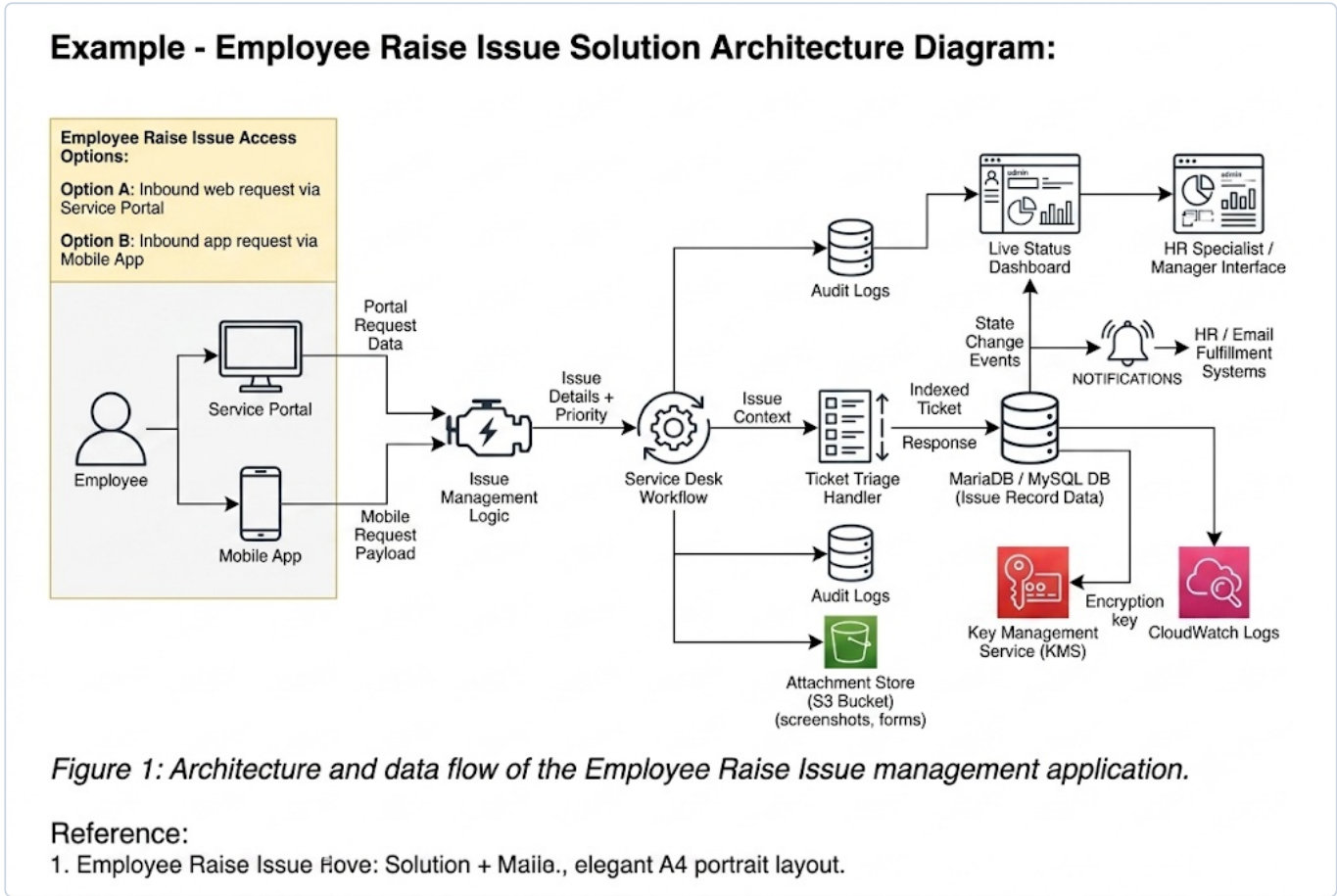


Figure 1: Architecture and data flow of the Employee Raise Issue management application.

Reference:

1. Employee Raise Issue Flove: Solution + Mailo., elegant A4 portrait layout.

SOLUTION ARCHITECTURE COMPONENT SPECIFICATIONS

Layer / Component	Description & Functionality	Technology / Platform
User Interface	Provides an intuitive presentation layer with custom Left and Right widgets for request creation and live status tracking.	Service Portal Framework, AngularJS, Bootstrap 3, HTML5, CSS3
Application Logic (Rules)	Evaluates client-side form behavior, enforces the Read-Only Ticket Number, and manages parent-child choice dependencies.	JavaScript (ES6), Client Scripts, ServiceNow UI Policies Engine
Data Ingestion	Maps front-end Record Producer form inputs directly into backend relational records with automated number assignment.	ServiceNow Record Producer Scripting, GlideRecord API
Scoped Application	Encapsulates all application artifacts, configurations, scripts, and widgets within an isolated scope to ensure security.	ServiceNow Scoped Application (Scope: x_1959081_emp_issue)
Data Storage	Custom relational data store engineered to hold issue records, categories, sub-categories, user references, and timestamps.	Custom Table (u_employee_raise_issue), ServiceNow Cloud DB Engine

SYSTEM CHARACTERISTICS & GOVERNANCE

Characteristic	Architectural Guarantee	Implementation
Security & Controls	Restricts administrative permissions while offering seamless end-user access. Protects ticket ID fields from user tampering.	ServiceNow ACLs, Application Scope Isolation, HTTPS/TLS 1.3
Scalability	3-tier cloud deployment capable of expanding across additional enterprise departments (HR, Facilities, IT, Legal).	ServiceNow Multi-Instance Cloud Architecture
Availability	Ensures 24/7 self-service portal accessibility without relying on manual support desk availability.	Advanced High Availability (AHA) Cloud Infrastructure

REFERENCE

<https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>